Beginner Guide to AKS Cluster Monitoring

Monitoring your Kubernetes cluster helps you spot container crashes and resource bottlenecks before your users experience downtime. Azure Monitor and Container Insights give you full visibility into pods and nodes without installing complicated third-party tools.

What you'll learn:

1. Turn On Container Insights
2. Detect Crashing Pods with KQL
3. Query Application Errors in Logs
4. Configure High Memory Alerts

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1

TIP 1 OF 4

Turn On Container Insights

Container Insights deploys a specialized monitoring agent across all nodes in your AKS cluster. This agent gathers performance metrics from the Kubernetes control plane, nodes, and running pods automatically. The data flows into an Azure Log Analytics workspace where it generates out-of-the-box performance graphs. You can toggle this on during cluster setup or enable it on an existing cluster using Azure CLI in just one step.

```
$az aks enable-addons -a monitoring -n myAKSCluster -g myResourceGroup --workspace-resource-id /subsc
```

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2

TIP 2 OF 4

Detect Crashing Pods with KQL

When containers misbehave, Kubernetes automatically restarts them until they fall into a CrashLoopBackOff state. Instead of running manual describe commands on dozens of pods, you can run a Kusto Query Language (KQL) query. Container Insights writes pod lifecycle changes into the KubePodInventory table in near real-time. This query filters for any pod restarting repeatedly so you can catch faulty code immediately.

```
KubePodInventory
 | where RestartCount > 3
 | project TimeGenerated, PodName = Name, Namespace, RestartCount, ContainerStatus
 | order by RestartCount desc
```

3

TIP 3 OF 4

Query Application Errors in Logs

Container Insights automatically scrapes all stdout and stderr console output and places it into the ContainerLogV2 table. This eliminates the need to grant cluster terminal access to developers who just need to inspect application errors. You can filter logs by specific namespaces, container names, or exact error text strings like 'Exception'. It speeds up root-cause analysis during high-stress production incidents.

```
ContainerLogV2
 | where PodNamespace == "production"
 | where LogMessage has "error" or LogMessage has "exception"
 | project TimeGenerated, PodName, LogMessage
 | take 50
```

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4

TIP 4 OF 4

Configure High Memory Alerts

Proactive alerts notify your team before out-of-memory (OOM) kills drop critical workloads. Using Azure Monitor metric alerts, you can watch node memory consumption over rolling 5-minute windows. If worker node memory exceeds your 80% threshold, Azure dispatches an automated notification. This gives your infrastructure team enough time to scale out the node pool before pods crash.

```
$az monitor metrics alert create \
   --name "Node-Memory-Alert" \
   --resource-group "myResourceGroup" \
   --scopes "/subscriptions/<sub-id>/resourceGroups/myRG/providers/Microsoft.ContainerService/managed
   --condition "avg node_memory_working_set_percentage > 80" \
   --window-size 5m --evaluation-frequency 1m
```

See Key Takeaways on next page >> linkedin.com/in/mramanrajsingh

Key Takeaways

+ Always configure CPU and memory requests/limits on every pod so monitoring metrics accurately reflect capacity.
+ Switch to the modern ContainerLogV2 schema to reduce ingestion costs and speed up your queries.
+ Scope KQL log queries to short time ranges (e.g. past 1 hour) to keep query responses fast.
+ Set budget alerts on your Log Analytics workspace to prevent accidental billing surprises.
+ Use Azure Action Groups to route critical alert notifications straight to Slack or Microsoft Teams.

Watch Out For

! Leaving debug-level logging enabled in production workloads, which inflates Log Analytics data ingestion bills.
! Relying solely on manual portal checks instead of setting up automated metric alerts for out-of-hours coverage.
! Forgetting to check node disk space, assuming cluster health only depends on CPU and memory.

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